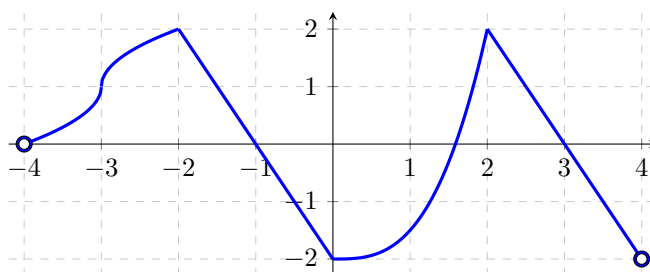


Problem 7

Problem 7

Problem 1 The graph of a function $g(x)$ with domain $(-4, 4)$ is given in the figure. This portions of this graph in the intervals $(-2, 0)$ and $(2, 4)$ are straight lines. The portion on the interval $(0, 2)$ is not parabolic.



For each of the limits below, give the form of the limit, then evaluate the limit.

(a) $\lim_{x \rightarrow -3} \frac{g(x) + \sin\left(\frac{\pi}{4}x\right)}{x\sqrt{x+4}}.$

Explanation. The numerator and denominator are each continuous at $x = -3$, and at $x = -3$ the denominator is nonzero. The entire fraction is continuous at $x = -3$. To find the limit, we can plug in the value. (We would call this form either “ $\frac{\#}{\#}$ ” or “Continuous”).

$$\begin{aligned} \lim_{x \rightarrow -3} \frac{g(x) + \sin\left(\frac{\pi}{4}x\right)}{x\sqrt{x+4}} &= \frac{g(-3) + \sin\left(\frac{-3\pi}{4}\right)}{-3\sqrt{-3+4}} \\ &= \frac{1 - \frac{1}{\sqrt{2}}}{-3}. \end{aligned}$$

(b) $\lim_{x \rightarrow 0^+} \frac{e^x}{|g(x) + 2|}.$

Explanation. $\lim_{x \rightarrow 0^+} e^x = e^0 = 1$ and $\lim_{x \rightarrow 0^+} |g(x) + 2| = 0$, so this limit has form $\frac{\#}{0}$. One-sided limits with form $\frac{\#}{0}$ are either $+\infty$ or $-\infty$, so we need to check the sign of the fraction.

Problem 7

We know the range of the famous function e^x is $(0, \infty)$, so the numerator here is always positive. The denominator is an absolute value, so it is not negative either. As long as x is near, but not equal, to 0, $g(x) + 2$ will be close, but not equal, to 0, so $|g(x) + 2|$ will be positive.

$$\lim_{x \rightarrow 0^+} \frac{e^x}{|g(x) + 2|} = \infty.$$

(c) $\lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x + g(x) - 3}$

Explanation. $\lim_{x \rightarrow 3} (x^2 + 2x - 15) = 0$ and $\lim_{x \rightarrow 3} (x + g(x) - 3) = 0$, so this limit has form $\frac{0}{0}$. In order to use algebra to simplify this fraction, we need a formula for $g(x)$ for x near 3. From the graph we see that in the interval $(2, 4)$, the graph of g is a straight line with slope -2 . The equation of that line is $y = -2x + 6$, so for x close to 3, we know $g(x) = -2x + 6$.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x + g(x) - 3} &= \lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x + (-2x + 6) - 3} \\ &= \lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{-x + 3} \\ &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 5)}{-(x - 3)} \\ &= \lim_{x \rightarrow 3} -(x + 5) \\ &= -8. \end{aligned}$$